Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	20 September 2026
Team ID	Individual
Project Name	Digital Metro Ticket Booking System using ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed metro ticket booking solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Ticket Booking Service Catalog**	• Create a Service Catalog item for metro ticket booking. • Allow passengers to select source, destination, passenger type and number of tickets.
FR-2	**Station & Journey Details**	• Maintain station details and map source/destination selections. • Validate required booking information before submission.
FR-3	**Automated Fare Calculation**	• Calculate the ticket fare automatically based on selected journey and passenger details. • Display the calculated fare clearly before confirmation.
FR-4	**Digital Payment & Booking**	• Support digital payment options such as UPI, cards and mobile wallets. • Record the booking and payment status after successful transaction.
FR-5	**QR Code Ticket Generation**	• Automatically generate a digital ticket with a QR code after successful booking. • Provide the passenger with access to the generated ticket.
FR-6	**Ticket Tracking & Notifications**	• Allow passengers to view booking/ticket details and status. • Provide relevant confirmation or ticket notifications after booking.

Non-functional Requirements:

Following are the non-functional requirements of the proposed metro ticket booking solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Service Catalog booking form should be simple, clear, mobile-friendly and easy for passengers to complete.
NFR-2	**Security**	Passenger and booking information should be protected, and payment and ticket-generation processes should follow appropriate access and security controls.
NFR-3	**Reliability**	Booking, fare calculation, payment status and QR-code generation should work consistently without losing ticket information.
NFR-4	**Performance**	Ticket booking, fare calculation and QR-code generation should respond quickly with minimal waiting time.
NFR-5	**Availability**	The digital ticket booking service should be available to passengers when they need to plan and book metro travel.
NFR-6	**Scalability**	The solution should support increasing passengers, bookings and transaction volumes without major redesign.